Innovation partner H2S

Media partner YOURSTORY

AI for Bharat Hackathon

Powered by aws

Team Name : ByteCoke

Team Leader Name : Tanisha Mathur

Problem Statement : Over ₹1.5 trillion in Indian government welfare remains unclaimed annually because low-literacy rural citizens face insurmountable linguistic, digital, and bureaucratic barriers to accessing their rights.

Brief about the Idea

The Gap:

Every year, ₹1.5 trillion in welfare benefits go unclaimed due to language barriers and middlemen.

The Solution:

Nyaaya.ai is a voice-first, offline AI assistant that empowers rural citizens to claim their rights with dignity.

Key Pillars

- **Voice-First:** Natural Hinglish NLU to understand complex life situations.
- **Offline-Always:** 100% functional on 2G/no-internet devices using local Llama 3.
- **Action-Ready:** Generates strategic week-by-week application plans.

Transforming welfare from a struggle into a right

How different is it from any of the other existing ideas?

- ✓ **Voice-First Hinglish:** While others use complex forms, we use natural speech understanding for low-literacy users.
- ✓ **100% Offline Core:** Unlike cloud-dependent apps, our Llama 3-powered engine works in villages with zero connectivity.
- ✓ **Action over Information:** We don't just list schemes; we provide a week-by-week strategy to maximize approval chances.

How will it be able to solve the problem?

- **Discovery:** Finds 3–5 eligible schemes per user vs. 0–1 on standard portals.
- **Verification:** On-device OCR reduces the 78% document-based rejection rate to <20%.
- **Dignity:** Reduces office visits by 75%, saving users ₹1,500+ in travel and agent fees.

USP of the proposed solution

- The **Community Success Moat:** A network of 100k+ verified peer success stories from the user's specific district, turning theoretical rules into proven outcomes.

Bridging the gap between government rights and rural reality through edge-AI.

List of Features Offered by the Solution

Voice-First Hinglish Interface

Uses natural speech-to-text to understand complex life situations in code-mixed Hindi/English, removing literacy barriers.

100% Offline Core

Fully functional on-device Llama 3 inference; no internet required for interviews, optimization, optimization, or OCR.

Strategic Scheme Optimizer

A constraint-based engine that generates week-by-week application timelines, prioritizing schemes by speed and benefit.

On-Device OCR Verifier

Auto-extracts details from Aadhaar and Ration cards using Google ML Kit to reduce document-based rejections from 78% to <20%.

Community Success Network

Semantic matching that connects users with verified peer success stories from their specific district.

Pro Mode for Volunteers

Bulk management dashboard allowing a single village volunteer to assist up to 50 elderly or illiterate beneficiaries.

Advanced AI at the edge, serving the last mile of Bharat.

Process Flow diagram or Use-case diagram

100% Offline Capability

Step 1: Voice Interview

User speaks naturally in Hindi/Hinglish; on-device Vosk STT converts audio to text (<300ms latency).

Step 2: Edge NLU

Local quantized Llama 3 8B extracts critical data points (age, income, location) without needing internet.

Step 3: Hardcoded Guardrail

Extracted data is validated against deterministic Pydantic rules to ensure 0% LLM hallucination on eligibility.

Step 4: Strategic Optimization

Google OR-Tools constraint solver generates the most efficient, week-by-week application timeline.

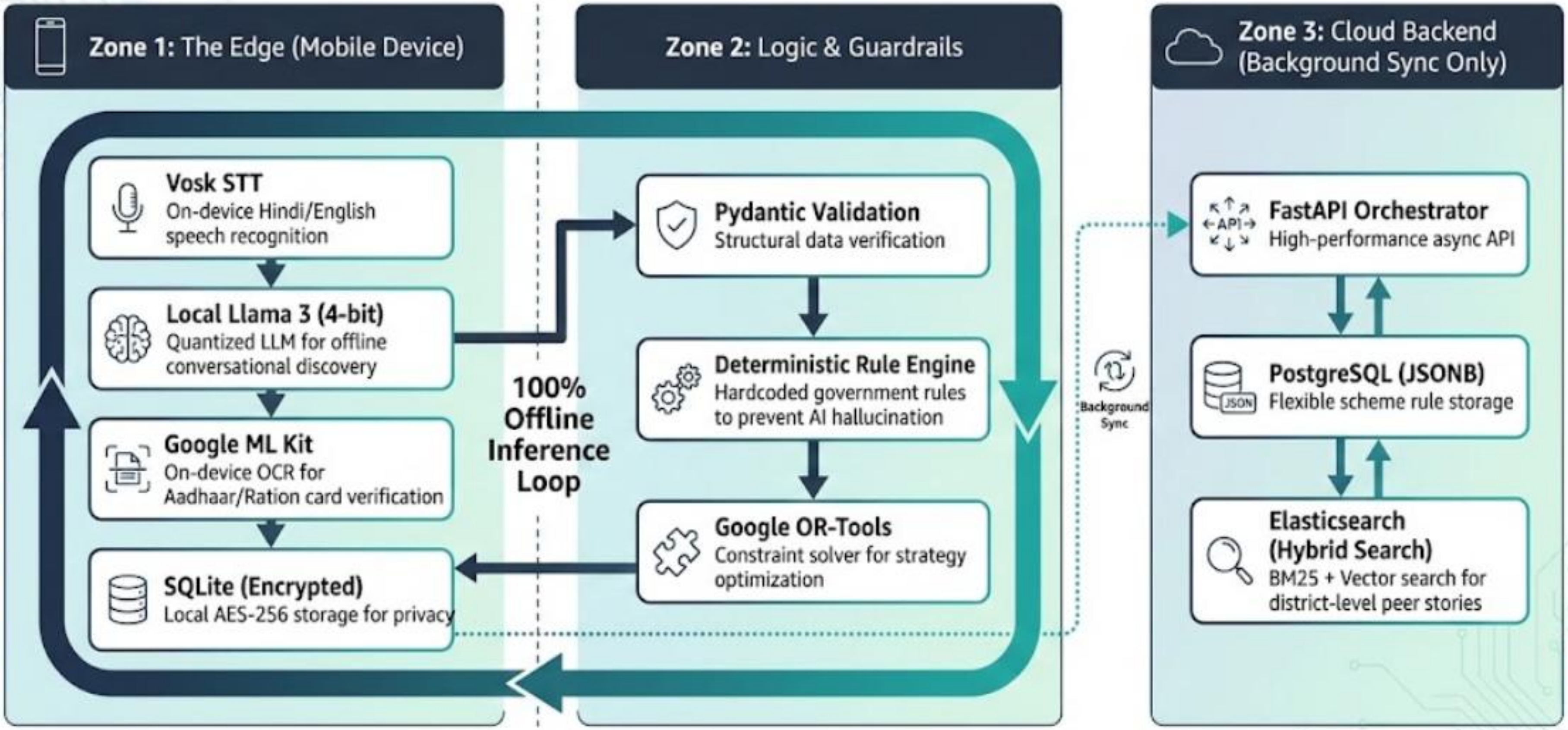
Step 5: Actionable Plan

System retrieves verified peer success stories and provides a final, printable document checklist for the user.

A seamless journey from conversational ambiguity to administrative success.

Wireframes/Mock diagrams of the proposed solution

Designed for the 'Next Billion' —prioritizing voice, clarity, and trust over complex forms.



Built for performance on ₹5,000 devices and reliability on 2G networks.

Technologies to be used in the solution:

Core AI & LLM

Llama 3 8B (4-bit quantized) for on-device, offline natural language understanding.

Voice Interface

Vosk Hindi Model (50MB) for ultra-low latency, on-device speech-to-text (<300ms).

Mobile Framework

Flutter for a smooth, high-performance UI experience on ₹5,000 Android devices.

Backend & Logic

FastAPI (Python) for asynchronous orchestration and Google OR-Tools for constraint-based strategy optimization.

Search & Retrieval

Elasticsearch utilizing Hybrid Search (BM25 + Semantic Vector) for precise district-level peer matching.

Data & Security

PostgreSQL for scheme rules and **SQLite** with **SQLCipher** (AES-256) for secure, on-device data persistence.

Computer Vision

Google ML Kit for privacy-preserving, on-device document OCR.

Leveraging cutting-edge Edge-AI and AWS infrastructure to deliver justice at scale.

Phase 1: Pilot & MVP (0–50k Users)

Engineering & Development

₹30L
(6 engineers over 6 months).

Data Collection & NGO Partnerships

₹40L
(Ground-verification of 100k success stories).

User Research & UX

₹10L
(Deep-field studies with rural beneficiaries).

Infrastructure (AWS)

₹50,000/month (Estimated for 2 API servers, PostgreSQL, and Redis clusters).

Scaling Milestone:

Phase 2 (200k Users)

Infrastructure cost scales to ~₹2L/month with auto-scaling and read replicas.

Phase 3 (1M Users)

Shift to a Federated Architecture to share infrastructure costs with NGO partners.

Strategic Note:

On-Device LLM Inference saves an estimated **₹25M/month** in cloud costs at a 10M-user scale.

A cost-efficient, high-impact model designed for long-term sustainability and government integration.

Innovation partner H2S | Media partner YOURSTORY

AI for Bharat Hackathon

Powered by aws

Thank You